

+84(039)326-8018 • mcu7uh@virginia.edu
Personal Website • Google Scholar • ORCID

CAREER INTERESTS

AI safety researcher focused on reliable, controllable, and accountable AI systems. I am especially interested in technical and policy-facing roles in **model policy, AI governance, safety evaluations, and reliable deployment** at frontier AI labs and responsible AI organizations.

- **Reliable Generative & Agentic Systems:** Developing methods for constrained generation, verification, robustness, and failure mitigation in high-stakes AI systems.
- **Responsible Deployment & Governance:** Translating technical safety research into deployable controls, evaluation protocols, and governance-relevant evidence for AI systems.

EDUCATION

University of Virginia Charlottesville, Virginia
Doctor of Philosophy, Computer Science *Aug 2026 - Present*
Research topic: Human-Agent Systems, Responsible Deployments, Constrained Optimization.

Beloit College Beloit, Wisconsin
Bachelor of Science, Mathematics *Aug 2021 – May 2025*
Relevant Coursework: Mathematical Statistics, Differential Equations, Complex Analysis, Topology, Algorithm Design and Analysis, Data Mining.

SKILLS

AI/ML: LLM evaluation, robustness analysis, contamination detection, constrained machine learning, graph learning, scientific ML.

Programming: Python, PyTorch, scikit-learn, NumPy, pandas, Git, LaTeX.

Research Methods: experimental design, statistical analysis, benchmark evaluation, technical writing.

EXPERIENCE

Vietnam Academy of Science and Technology Hanoi, Vietnam
Researcher, Institute of Information Technology *May 2025 – May 2026*

- Research methods to make generative systems more reliable, with a focus on robustness, constraint-aware, and detecting hidden failures in large language models.
- Studied how small changes in structured data can mitigate model behavior, helping build tools that make AI pipelines more stable and easier to evaluate.
- Co-designed **Residual Noise-Fingerprinting (RN-F)**, a method for detecting whether language models may have been exposed to test data during training, supporting more trustworthy AI evaluation.

Massachusetts Institute of Technology Cambridge, Massachusetts
Research Intern, MIT Summer Research Program *June 2024 – August 2024*

- Built regression models to support safer environmental decision-making at a nuclear waste disposal site.
- Enforced physical constraints to make predictions more reliable when data conditions changed, reducing the risk of misleading model outputs in a high-stakes scientific setting.

PUBLICATIONS

1. **Le, Vu Anh** and Dik, Mehmet, "A Mathematical Analysis of Neural Operator Behaviors," Chapter 23 in *Advances in Quantum Calculus and Functional Analysis*, Taylor & Francis Group, July 2025
2. **Le, Vu Anh**, Nguyen, Dinh Duc Nha, Nguyen, Phi Long, and Sood, Keshav, "RN-F: A Novel Approach for Mitigating Contaminated Data in Large Language Models," in *International Conference on Machine Learning Workshop on Data in Generative Models*, June 2025

SELECTED PRESENTATIONS

1. "RN-F: A Novel Approach for Mitigating Contaminated Data in Large Language Models."
 - Workshop on Data in Generative Models, *International Conference on Machine Learning*, July 2025
2. "Mathematical Foundations of Neural Operators."
 - Infinite Possibilities Conference, *University of Chicago*, April 2025
 - National Conference on Undergraduate Research 2025, *Pittsburgh*, April 2025

SELECTED AWARDS AND HONORS

UVA Computer Science Scholar , University of Virginia, Full PhD sponsorship	<i>Aug 2026 – May 2031</i>
Cum Laude , Beloit College	<i>May 2025</i>
Presidential Scholarship , Beloit College, Awards 48,000 USD annually	<i>Aug 2021 - May 2025</i>
Board of Trustees Grant , Beloit College, Awards 10,000 USD annually	<i>Aug 2021 - May 2025</i>
Dean's List , Beloit College	<i>Every semester</i>
MIT Summer Research Program , Massachusetts Institute of Technology, Fully funded	<i>June 2024</i>
Weissberg Human Rights Grant , Weissberg Foundation, Awards 1,000 USD	<i>March 2024</i>
Semifinalist , InSPiR2eS Global Pitching Research Competition 2023 (IGPRC 2023)	<i>Jan 2024</i>
Station1 Frontiers Fellowship , Station1, Awards 13,500 USD	<i>June 2023</i>

REFERENCES

Prof. Ferdinando Fioretto

Associate Professor of Computer Science
Department of Computer Science
University of Virginia
Email: fioretto@virginia.edu; Phone number: +1 (434) 924-3072

Prof. Nguyen Viet Anh

Associate Professor and President of Scientific Council
Institute of Information Technology
Vietnam Academy of Science and Technology
Email: anhnv@ioit.ac.vn; Phone number: +84 (091) 762-6028

Prof. Mehmet Dik

Chair and Professor
Department of Mathematics, Computer Science, and Physics
Rockford University
Email: mdik@rockford.edu; Phone number: +1 (815) 226-4135

Prof. Haruko Wainwright

Assistant Professor
Department of Nuclear Science and Engineering
Massachusetts Institute of Technology
Email: hmwainw@mit.edu; Phone number: +1 (617) 324-4395